Reference and Referential Relations

Reference • Co-reference • Endophora • Exophora • Deixis

Anaphora • Cataphora • Demonstratives • Tense • Pronominals

1. What is Reference?

Reference is the relationship between a linguistic expression and the entity, person, object, place, event or situation that the expression is used to identify.

The entity identified is often called the referent.

Reference is interpreted using linguistic context, the physical situation, shared knowledge and discourse context.

Example: “Riya bought a book. The book was expensive.” → “the book” refers to the particular book Riya bought.

Important: reference is a relation between an expression and what it identifies in a context—not simply the dictionary meaning of a word.

Reference: Expression and Referent

A referring expression can be a proper name, noun phrase, pronoun, demonstrative or other expression.

Example: “Dr. Mehta arrived.” → “Dr. Mehta” identifies a particular person.

Example: “She arrived.” → “she” identifies a person recoverable from context.

Example: “That is my bag.” → “that” identifies an entity salient in the situation.

The same expression can refer to different entities in different contexts: “I” refers to the current speaker; “here” refers to the relevant location.

2. Reference is Context-Dependent

Compare: “I am tired.” The referent of “I” changes with the speaker.

“Put it there.” The referents of “it” and “there” depend on the situation.

“Tomorrow we will meet.” The interpretation of “tomorrow” depends on the time of utterance.

Thus, reference can depend on discourse context as well as the physical context of speaking.

This context sensitivity is especially important for deixis.

3. Co-reference

Co-reference occurs when two or more linguistic expressions refer to the same entity.

Example: “Neha entered the room. She sat near the window.” → “Neha” and “she” are co-referential.

Example: “The Prime Minister addressed the nation. The leader spoke for ten minutes.” → the two expressions may be co-referential in the relevant context.

Co-reference concerns identity of referent; anaphora concerns a particular discourse relationship between expressions.

Not every co-referential pair is necessarily anaphoric.

Co-reference vs. Anaphora

Co-reference: two expressions have the same referent.

Anaphora: an expression is interpreted by connecting it to an antecedent in the discourse.

Example: “Arun bought a car. He likes it.” → “he” is anaphoric to “Arun”; “it” is anaphoric to “a car”; the pairs are also co-referential.

Example: “The winner is Dr. Rao. Dr. Rao will receive the award.” → the repeated name is co-referential, but it is not a typical pronoun-anaphor relationship.

Therefore, co-reference is about shared reference; anaphora is about discourse linkage.

4. Endophora

Endophora is reference to something inside the linguistic/discourse context.

The interpretation of an endophoric expression can be recovered from another expression in the discourse.

Example: “I bought a new phone. It has a good camera.” → “it” refers to “a new phone” inside the discourse.

Endophora is commonly divided into anaphora and cataphora.

Memory aid: endo = inside; the relevant information is inside the discourse.

5. Exophora

Exophora is reference to something outside the linguistic text/discourse.

The referent is identified from the surrounding situation, shared knowledge or perceptual context.

Example: “Look at that!” → “that” refers to something salient in the situation, not necessarily mentioned in the discourse.

Example: “Please close the door.” → “the door” can refer to a physically present door.

Memory aid: exo = outside; the referent is supplied by context beyond the words already spoken.

6. Types of Endophora

ANAPHORA: backward-looking reference; an expression refers to something already mentioned.

Example: “Ravi bought a laptop. He uses it every day.”

CATAPHORA: forward-looking reference; an expression points ahead to an expression introduced later.

Example: “When he entered the room, Ravi smiled.” → “he” is interpreted through the later expression “Ravi”.

Both create links across discourse, but the direction of interpretation differs.

7. Anaphora

Anaphora is a relation in which an expression (the anaphor) depends on an earlier expression (the antecedent) for its interpretation.

Example: “Priya lost her keys. She searched everywhere.” → “She” = Priya.

Example: “I read the article and discussed it with my teacher.” → “it” = the article.

The antecedent may be a noun phrase, clause or larger discourse segment.

Anaphora contributes to cohesion by allowing speakers/writers to avoid unnecessary repetition.

Anaphora: More Examples

Personal pronoun: “Rahul arrived. He sat down.”

Demonstrative: “The company announced a merger. This surprised investors.”

Definite noun phrase: “A dog entered the park. The animal looked frightened.”

Possessive: “Maya bought a new laptop. Her laptop is lightweight.”

Anaphoric interpretation requires discourse context and sometimes world knowledge.

8. Cataphora

Cataphora is a relation in which an expression is interpreted through a later expression.

Example: “Before she spoke, Ananya took a deep breath.” → “she” = Ananya.

Example: “If you see him, tell Raj that I called.” → “him” can be interpreted from context; cataphoric interpretation is strongest when the later expression supplies the identity.

Cataphora can create suspense, emphasis or a particular information structure.

Cataphora: More Examples

“Although he was tired, Professor Singh continued the lecture.” → “he” = Professor Singh.

“When she arrived, Maria went straight to the office.” → “she” = Maria.

“This is what I mean: the experiment needs more data.” → “this” anticipates the following proposition.

Cataphora can operate with pronouns and demonstratives, and can also point toward later clauses or discourse units.

9. Deixis

Deixis is reference whose interpretation depends strongly on the context of utterance, especially the speaker, addressee, time and place.

A deictic expression ‘points’ to a referent from the speech situation.

Classic examples: I, you, here, there, now, then, today, tomorrow, this, that.

The referent of a deictic expression can shift when the speaker, location or time changes.

Deixis is therefore closely associated with exophoric reference, although some deictic expressions can also have discourse-anaphoric uses.

Deixis: The Deictic Centre

Interpretation is often organized around a deictic centre.

PERSON: who is speaking and who is being addressed?

SPACE: where is the relevant location relative to the deictic centre?

TIME: when is the utterance produced relative to the deictic centre?

SOCIAL: what social relationship or status is encoded in the choice of expression?

10. Person Deixis

Person deixis locates participants relative to the speech event.

First person: I, me, we, us → speaker (and possibly associated group).

Second person: you → addressee(s).

Third person: he, she, they → entities not functioning as speaker/addressee in the speech event.

Example: “I will meet you here tomorrow.” The referents of I and you depend on the participants in the speech event.

“I” does not have one permanent referent; it refers to whoever is the current speaker.

“You” refers to the current addressee(s), and its referent changes across conversations.

“We” can be inclusive or exclusive depending on language/context: speaker + addressee(s), or speaker + others excluding addressee(s).

Third-person forms can also be used in ways sensitive to discourse perspective.

Person deixis is a clear example of context-dependent reference.

11. Spatial Deixis

Spatial deixis identifies locations or objects relative to a spatial centre.

Common expressions: here, there, this, that, these, those.

“Here” generally identifies a location associated with the speaker.

“There” generally identifies a location away from the deictic centre.

Example: “Put these books here and those books there.” Interpretation depends on the physical/discourse situation.

Demonstratives

Demonstratives are expressions that direct attention to an entity, location or discourse content.

English: this/these typically contrast with that/those in terms of proximity, though actual use is more complex.

Spatial/deictic use: “This chair is comfortable.”

Anaphoric/discourse use: “He missed the train. This caused a delay.”

Cataphoric/discourse use: “This is the problem: we have no evidence.”

Thus, demonstratives can be deictic, anaphoric or cataphoric depending on context and construction.

12. Temporal Deixis

Temporal deixis locates events or times relative to the time of speaking or another temporal reference point.

Examples: now, then, today, yesterday, tomorrow, next week, last year.

“I am busy now.” → “now” is anchored to the utterance time.

“I met her yesterday.” → “yesterday” is calculated relative to the utterance time.

Temporal expressions can shift when the utterance time changes.

Tense and Temporal Reference

Grammatical tense can establish temporal relations involving the speech time.

Present: “She works in Delhi.” → typically presents the situation as current or generally valid.

Past: “She worked in Delhi.” → places the situation before the relevant reference time.

Future constructions: “She will work in Delhi.” → locate the situation after the relevant reference point.

Tense is not identical to deixis, but tense often has a deictic component because temporal interpretation is anchored to speech time or another contextually supplied reference time.

13. Social Deixis

Social deixis encodes information about social relationships, status, politeness, familiarity or relative social position.

English has limited grammatical social deixis compared with languages that have elaborate honorific systems.

Examples include titles and address forms such as Professor, Doctor, Sir/Madam, and honorific forms in languages such as Hindi, Japanese and Korean.

Social deixis shows that referring to a person can also communicate how the speaker positions the relationship socially.

14. Pronominals

Pronominals are expressions such as personal pronouns that can stand in place of fuller noun phrases.

Examples: I, you, he, she, it, we, they; their corresponding object/possessive forms also participate in reference.

Deictic use: “I am here.” → I depends on the speaker; here depends on location.

Anaphoric use: “Meera arrived. She sat down.” → she refers back to Meera.

Cataphoric use: “When he arrived, Arjun sat down.” → he refers forward to Arjun.

Thus, the same grammatical category can participate in different reference relations.

Pronominals: Deixis vs. Anaphora vs. Cataphora

DEICTIC: “I need this.” → interpretation depends on the speech situation.

ANAPHORIC: “Ravi came in. He sat down.” → pronoun points back to Ravi.

CATAPHORIC: “Before she spoke, Anita checked the microphone.” → pronoun points forward to Anita.

The form “he/she/this” alone does not determine the relation; context and discourse structure do.

This is why reference is best understood as an interaction between linguistic form and context.

15. One Expression, Different Referential Functions

“this” can be deictic: “This book is mine.” (object present/indicated)

“this” can be anaphoric: “He failed the exam. This upset him.” (refers to preceding situation)

“this” can be cataphoric: “This is the issue: the data are incomplete.” (points to following content)

“she” can be deictic in a context where the person is identifiable from the situation.

“she” can be anaphoric when an antecedent occurs earlier, or cataphoric when an identifying expression occurs later.

17. Ambiguity in Reference

A referring expression may have more than one plausible antecedent.

Example: “Ravi told Amit that he was late.” → Who does “he” refer to?

Interpretation may use grammatical constraints, semantic compatibility, discourse focus and world knowledge.

Some references remain ambiguous without additional context.

Therefore, reference resolution is not always a simple word-matching task.

19. Summary Table

REFERENCE: expression identifies an entity/situation in context.

CO-REFERENCE: two or more expressions identify the same entity.

ENDOPHORA: reference interpreted through discourse-internal material.

EXOPHORA: reference depends on context outside the discourse.

ANAPHORA: backward-looking endophoric relation.

CATAPHORA: forward-looking endophoric relation.

DEIXIS: context-dependent pointing; major types include person, spatial, temporal and social deixis.

Key Takeaways

Reference links linguistic expressions with entities or situations.

Co-reference means shared referent.

Endophora looks inside discourse; exophora depends on context outside the discourse.

Anaphora points backward; cataphora points forward.

Deixis anchors reference to the speech situation.

Person, spatial, temporal and social deixis encode different dimensions of context.

Demonstratives, tense and pronominals can participate in deictic, anaphoric and cataphoric interpretation.

20. Integrated Example

“Yesterday, I met Professor Rao here. She gave me this book. I will read it tomorrow.”

“I” → person deixis; “yesterday/tomorrow” → temporal deixis; “here” → spatial deixis.

“She” → anaphoric reference to Professor Rao.

“this book” → demonstrative; may be deictic if the book is physically indicated, while “this” can also have discourse-related uses.

“it” → anaphoric reference to the book.

The example shows how several reference mechanisms operate simultaneously.